

# PRANJAL PARMAR

M.Sc. Computer Science Student | GenAI | Agentic AI | Computer Vision | NLP | RAG

@pranjalp@mscloud.uni-paderborn.de +49-17643390704 Paderborn, Germany linkedin github

## EXPERIENCE

### Werkstudent Data Annotator

Metamorphosis GmbH (20h/week)

- Feb 2024 – Present Paderborn, Germany
- Annotated and validated high-precision medical imaging datasets for AI-assisted surgery.
  - Optimized annotation workflows to accelerate the delivery of model-ready datasets.
  - Enforcing strict clinical quality standards to ensure reliability in medical AI decision-making.

### Business Analyst

Mapup Inc. — Reference Letter

- Feb 2022 – Nov 2022 India
- Analyzed the problems in the transportation, OEMs, Automotive, First and Last-Mile industry
  - Understood the needs of stakeholders and translated them into technical requirements for product development.
  - Developed KPI dashboards providing real-time operational decision support.

## EDUCATION

### M.Sc. in Computer Science

Paderborn University

- Apr 2023 – Present Paderborn, Germany
- GPA: 2.3** | Focus: Data and Intelligence
  - Master's Thesis (Fraunhofer IEM, Heylexie.de)** May '26 – Oct '26
    - Title:** Design and Evaluation of a Neurosymbolic Child-Safe AI Architecture for Children Aged 4-7
    - Research Objective:** Proposing a "Safety-by-Design" AI framework for children aged 4-7 through a novel hybrid neurosymbolic architecture.
    - Implementation:** Integrating a Neural Semantic Parser with a Symbolic Safety Governor for deterministic, traceable, and explainable safety control.
    - Evaluation:** Benchmarking against NVIDIA NeMo Guardrails using age-specific interaction datasets.

**Master's Project: GenSys (Fraunhofer IEM, KI-Markplatz)** Oct '23 – Aug '24

- Automated SysML 2.0 diagram generation, modification and explanation from natural language prompts and images using GPT-4 Vision.
- Developed a Streamlit interface to streamline system modeling for engineering teams.

### B.E. in Computer Engineering

Gujarat Technological University

- 2018 – 2022 India
- GPA: 1.3**

## PASSION PROJECTS

### VisionExpert-RAG

Built a visual RAG system using ColPali and Mistral-7B for technical document auditing.

### ACC-RAG Architecture

Developed an Adaptive Contextual-Retriever for local systems, fusing text with OCR diagram labels to extract all the details from engineering manuals.

### Multimodal Fracture Detection

Developed a CNN-based system using MedGemma for image-grounded explanations.

### Brain Tumor Detection with YOLO

Engineered a YOLO-based localization model using PyTorch and OpenCV.

### Handwritten Digit Recognizer

Built an automated CI/CD pipeline that trains and deploys MNIST digit classification models.

## DOMAIN

- Machine Learning
- GenAI
- Agentic AI
- Computer Vision
- Neurosymbolic AI

## SKILLS

- Languages** Python, R, Java, SQL, LaTeX
- AI/ML:** PyTorch | LangChain | NeMo Guardrails | YOLO | CNN | RAG | SysML
- Tools:** Docker | Git | FastAPI | Streamlit | Power BI | OpenCV | Pandas | CI/CD

## LANGUAGES

English (Fluent)  
German (A2 Level and Learning)  
Hindi / Gujarati (Native)

## CERTIFICATES

- Data Analytics Professional** | Google
- Python for Everybody** | Michigan University
- Data Analysis with Python** | Jovian.ml

## VOLUNTEERING

- Member, **COLOURS EU Alliance**
- Member, **AIESEC**
- Volunteer, **ASV Paderborn**

## INTERESTS

- Community Building
- Public Speaking
- Table Tennis
- Volleyball
- Cycling